KALEY BRAUER

NSF Astrophysics Postdoctoral Fellow
Institute for Theory and Computation, Harvard University
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EDUCATION

Ph.D. in Physics 2023
Massachusetts Institute of Technology
Thesis: *The Legacy of the First Galaxies*
GPA: 5.0/5.0

Sc.B. in Physics 2017
Brown University
University honors, departmental honors, GPA: 4.0/4.0

ACADEMIC POSITIONS

AI Alignment Research Fellow 2026 – present
Cambridge Boston Alignment Initiative; Anthropic

NSF Astrophysics Postdoctoral Fellow 2023 – present
Harvard University

US Department of Energy Graduate Fellow; Graduate Researcher 2017 – 2023
Massachusetts Institute of Technology

Visiting Research Fellow Fall 2021
Lawrence Berkeley National Laboratory

HONORS & AWARDS

Research Corporation for Science Advancement Fellow	2025
NSF Astronomy & Astrophysics Postdoctoral Fellowship ($330K; ~8 awarded annually)	2023
Spot Awards, MIT, *in recognition of community service to School of Science*	2020 & 2022
DOE Computational Science Graduate Fellowship (~$450K; ~25 awarded annually)	2018
NSF Graduate Research Fellowship ($\sim \$150,000$)	2018
MIT Whiteman Fellowship (~$100K)	2017
R. Bruce Lindsay Award, Brown University, *given to senior for excellence in physics*	2017
Eva A. Mooar Prize, Brown University, *given to woman for academic excellence*	2017
Sigma Xi Research Honor Society	2017
Karen T. Romer Undergraduate Teaching and Research Award, Brown University	2015

PUBLICATIONS

Refereed

Journal Articles

16. Cannon, R. E., Rozek, A., **Brauer, K.**, Busch, M. W., Snodgrass, C., Benner, L. A. M., Brozovic, M., Howell, E., Nolan, M. C., Rabus, M., Sajadian, S., Springmann, A., Taylor, P., Zambrano-Marin, L. F. (2026). The shape and spin state of (275677) 2000 RS11 from ground-based radar and optical observations. *in press at the Monthly Notices of the Royal Astronomical Society.*

15. Andersson, E. P., Rey, M. P., Yates, R. M., Read, J. I., Agertz, O., Ji, A. P., Mead, J., **Brauer, K.**, Mac Low, M. (2026). EDGE-INFERNO: The impact of chemical enrichment assumptions on the individual stars of a simulated faint dwarf galaxy. *in press at The Astrophysical Journal*. arXiv:2511.05695

14. Hazlett, R., Mead, J., Visbal, E., Bryan, G., Mac Low, M., Kulkarni, M., Andersson, E. P., **Brauer, K.**, Wise, J. (2025). From Primordial Stars to Early Galaxies: A Semi-Analytic Model Calibrated with AEOS and Renaissance. *submitted to The Astrophysical Journal*. arXiv:2510.11629

13. Mead, J., **Brauer, K.**, Bryan, G. L., Mac Low, M. M., Ji, A. P., Wise, J. H., Andersson, E. P., Frebel, A., Emerick, A., Cote, B. (2025). AEOS is Mixing it Up: The (In)homogeneity of Metal Mixing Following Population III Star Formation. *submitted to The Astrophysical Journal*. arXiv:2509.13580

12. **Brauer, K.**, Mead, J., Ji, A. P., Wise, J. H., Bryan, G. L., Mac Low, M. M., Emerick, A., Andersson, E. P., Frebel, A., Cote, B. (2025). AEOS: The Impact of Pop III Initial Mass Function and Star-by-Star Models in Galaxy Simulations. *The Astrophysical Journal*, 993, 2.

11. Mead, J., **Brauer, K.**, Bryan, G. L., Mac Low, M. M., Ji, A. P., Wise, J. H., Emerick, A., Andersson, E. P., Frebel, A., Cote, B. (2025). AEOS: Transport of metals from minihalos following Population III stellar feedback. *The Astrophysical Journal*, 980, 62.

10. **Brauer, K.**, Emerick, A., Mead, J., Ji, A. P., Wise, J. H., Bryan, G. L., Mac Low, M. M., Cote, B., Andersson, E. P., Frebel, A. (2025). AEOS: Star-by-Star Cosmological Simulations of Early Chemical Enrichment and Galaxy Formation. *The Astrophysical Journal*, 980, 41.

9. Chiti, A., Mardini, M. K., Limberg, G., Frebel, A., Ji, A. P., Reggiani, H., Ferguson, P., Andales, H. D., **Brauer, K.**, Li, T. S., Simon, J. D. (2024). Signatures of Extragalactic First Stars in the Large Magellanic Cloud. *Nature Astronomy*, 8, 637647.

8. **Brauer, K.**, Andales, H., Ji, A. P., Mardini, M., Frebel, A., Gomez, F. A., & O'Shea, B. W. (2022). Possibilities and Limitations of Kinematically Identifying Stars from Accreted Ultra-Faint Dwarf Galaxies. *The Astrophysical Journal*, 937, 14.

7. Ji, A. P., Naidu, R., **Brauer, K.**, Ting, Y., & Simon, J. (2022). Chemical Abundances of the Typhon Stellar Stream. *Monthly Notices of the Royal Astronomical Society*, 519, 4467-4478.

6. Mardini, M., Frebel, A., Chiti, A., Meiron, Y., **Brauer, K.**, Ou, X. (2022). Characterization of the Metal Weak Thick Disk of the Milky Way. *The Astrophysical Journal*, 936, 78.

5. **Brauer, K.**, Ji, A. P., Drout, M., Frebel, A. (2021). Collapsar R-Process Yields Can Reproduce [Eu/Fe] Abundance Scatter in Metal-Poor Stars. *The Astrophysical Journal*, 915, 81.

4. Gull, M., Frebel, A., Hinojosa, K., Roederer, I. U., Ji, A. P., **Brauer, K.** (2021). R-process-rich stellar streams in the Milky Way. *The Astrophysical Journal*, 912, 52.

3. **Brauer, K.**, Ji, A. P., Frebel, A., Dooley, G. A., Gomez, F. A., & O'Shea, B. W. (2019). The Origin of r-process Enhanced Metal-Poor Halo Stars In Now-Destroyed Ultra-Faint Dwarf Galaxies. *The Astrophysical Journal*, 871, 247.

2. **Brauer, K.**, Vrtilek, S. D., Peris, C., & McCollough, M. (2018). Phase-resolved spectroscopy of the low-mass X-ray binary V801 Ara. *Monthly Notices of the Royal Astronomical Society*, 478, 4894-4904.

Refereed Workshop Papers

1. **Brauer, K.**, Dash, A. P., Vyas, M. J., Salim, A. (2025). From Simulations to Surveys: Domain Adaptation for Galaxy Observations. *Machine Learning and the Physical Sciences (ML4PS) Workshop at NeurIPS 2025.* arXiv:2511.18590

Non-Refereed

2. **Brauer, K.** (2021). "I'll Finish it This Week" And Other Lies. *arXiv April Fools Paper.* arXiv:2103.16574

1. **Brauer, K.**, Ji, A., Hattori, K., Escobar, S., & Frebel, A. (2019). Kinematics of highly r-process-enhanced halo stars: Evidence for origins in now-destroyed ultra-faint dwarf galaxies. *Proceedings of the International Astronomical Union*, 14(S353), 71-74.

PRESENTATIONS

INVITED

AstroAI Workshop Talk at Harvard	Cambridge, MA, USA; Jun 2026
Astrophysics Theory Seminar Talk at UC San Diego	San Diego, CA, USA; Feb 2026
Talk at From Galaxies to Planets: An Elemental Journey	Ringberg Castle, Germany; Feb 2026
Colloquium Talk at Rensselaer Polytechnic Institute (RPI)	Troy, NY, USA; Oct 2025
Astronomy & Data Science Talk at Yale University	New Haven, CT, USA; May 2025
Astronomy Seminar Talk at Columbia University	New York, NY, USA; Mar 2025
Fundamental Physics Seminar Talk at Brown University	Providence, RI, USA; Mar 2025
Astronomy Seminar Talk at Lund University	Lund, Sweden; Mar 2025
Astronomy Seminar Talk at American Museum of Natural History	New York, NY, USA; Oct 2024
Nuclear Astrophysics Seminar Talk at IReNA	East Lansing, MI, USA; Sep 2024
Colloquium Summer Talk at Harvard-Smithsonian CfA	Cambridge, MA, USA; July 2023
Talk at US Department of Energy CSGF Program Review	Washington, DC, USA; July 2022
High Performance Computing Talks at IHPCSS	Athens, Greece; June 2022
Colloquium Talk at University of Melbourne	Melbourne, Australia; Oct 2021
Seminar Talk at Computational Research in Boston and Beyond	Boston, USA; Oct 2021
Seminar Talk at Carnegie Observatories	Pasadena, CA, USA; Oct 2020

CONTRIBUTED

Talk at NSF Postdoctoral Fellows Symposium 2026	Phoenix AZ, USA; Jan 2026
Poster at NeurIPS 2025 ML4PS Workshop	San Diego, CA, USA; Dec 2025
Talk at IAU Symposium 403 on Galactic Outskirts	Cordoba, Spain; Oct 2025
Poster at Tracing Cosmic Evolution with Clusters of Galaxies V	Sesto, Italy; July 2025
Talk at Galactic Frontiers II at Dartmouth	Hanover, CT, USA; Jun 2025
Talk at First Galaxies Meeting at Oxford	Oxford, UK; Apr 2025
Talk at NSF Postdoctoral Fellows Symposium 2025	Washington DC, USA; Jan 2025
Talk at IAU Symposium 395 on Stellar Populations	Paraty, Brazil; Nov 2024

Talk at Astrophysical Origins of Carbon	Tokyo, Japan; Sep 2024
Talk at First Stars VII at CCA	NYC, NY, USA; May 2024
Talk at Galaxies from Scratch 2024	Vienna, Austria; Feb 2024
Talk at 243rd Meeting of the American Astronomical Society	New Orleans, LA, USA; Jan 2024
Talk at NSF Postdoctoral Fellows Symposium 2024	New Orleans, LA, USA; Jan 2024
Talk at Flatiron CCA Galactic Frontiers	NYC, NY, USA; July 2023
Talk at IAU Symposium 377 on Early Disk Galaxy Formation	Kuala Lumpur, Malaysia; Feb 2023
Talk at 241st Meeting of the American Astronomical Society	Seattle, WA, USA; Jan 2023
Talk at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	South Bend, IN, USA; May 2022
Talk at 2021 GALAH Science Meeting	Sydney, Australia; June 2021
Talk at *Linking the Galactic and Extragalactic*	Wollongong, Australia; Dec 2020
Talk at 235th Meeting of the American Astronomical Society	Honolulu, HI, USA; Jan 2020
Talk at IAU Symposium 353 on Galactic Dynamics	Shanghai, China; July 2019
Poster at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	East Lansing, MI, USA; May 2019
Talk at JINA-CEE Frontiers in Nuclear Astrophysics Meeting	South Bend, IN, USA; May 2018
Poster at 229th Meeting of the American Astronomical Society	Grapevine, TX, USA; Jan 2017
Poster at 47th Meeting of the Division of Planetary Science	Washington, DC, USA; Nov 2015
Talk at SETI Institute's Summer Colloquium Series	Mountain View, CA, USA; Aug 2015

TECHNICAL SKILLS

Programming Python, C/C++, MPI, OpenMP, GPU acceleration, Slurm, Git
Machine Learning PyTorch, TensorFlow, HuggingFace, mechanistic interpretability

TEACHING

8.03 Vibrations and Waves Teaching Assistant — Spring 2023
Physics Department, Massachusetts Institute of Technology

8.S30 Stellar Archaeology Teaching Assistant — Fall 2022
Physics Department, Massachusetts Institute of Technology

High Performance Computing Summer School — Summer 2022
International High Performance Computing Summer School; Athens, Greece

Cosmology Course Designer and Instructor — Spring 2021
Spark, MIT Educational Studies Program

***Catalyst* Computer Science Instructor** — Spring 2019
Citizen Schools and Mass STEM Hub

PHYS0220/0270 Astronomy Teaching Assistant — Fall 2014 - Spring 2015
Physics Department, Brown University

PHYS0030 Introductory Physics Workshop Assistant — Fall 2014
Physics Department, Brown University

LEADERSHIP & SERVICE

Leader & Mentor to Astronomy Graduate Students 2023 - present
Astronomy Mentorship Program for Upcoming Postdocs (AMP-UP)

Subject-matter expert reviewer in NASA peer review 2025 - present
NASA

Crew Scientist for Analog Astronaut Mission Nov - Dec 2025
Mars Desert Research Station

Pi Day Outreach Organizer 2024 - 2025
Harvard University

Graduate Women in Physics Leader 2019 - 2023
Massachusetts Institute of Technology

Reviewer for JOSS 2022 - present
Journal of Open Source Software

President, MIT Salsa Club 2018 - 2022
Massachusetts Institute of Technology

Physics Representative, Diversity and Inclusion Committee 2018 - 2022
MIT Graduate Student Council

Treasurer, Students for the Exploration and Development of Space 2017 - 2021
Massachusetts Institute of Technology

Adopt-a-Physicist Volunteer 2018 - 2020
Sigma Pi Sigma

Observer & Designer, MIT Sidewalk Astrogazers 2017 - 2019
MIT Kavli Institute for Astrophysics and Space Research

Head of Design Team, The Triple Helix Magazine 2014 - 2017
Brown University

Women in Physics Co-coordinator 2016 - 2017
Brown University

Physics Show Presenter Summer 2014
Physics & Astronomy Department, Texas A&M University